

# MAT 4233 — Weekly Schedule

Modern Abstract Algebra · Fall 2026 · UT San Antonio

## Texts

### Primary text

- **KvP** — Kramer & von Pippich, *From Natural Numbers to Quaternions*
- **Beardon** — Beardon, *Algebra and Geometry*

### Supplementary reading

- **Needham** — Needham, *Visual Complex Analysis, 25th Anniversary Edition*

### Handouts & papers

- **Conrad** — Conrad, *The Gaussian Integers*
- **Arnold–Rogness** — Arnold & Rogness, *Möbius Transformations Revealed*

## Schedule

| Date       |                | Topic                                                              | Reading              | Due           |
|------------|----------------|--------------------------------------------------------------------|----------------------|---------------|
| Thu 20 Aug |                | Course launch; divisibility and the division algorithm             | KvP §1.1-1.2         |               |
| Tue 25 Aug |                | Primes, the Fundamental Theorem of Arithmetic, gcd and lcm         | KvP §1.3-1.5         |               |
| Thu 27 Aug |                | Semigroups, monoids, groups — all of them $\mathbb{Z}/n\mathbb{Z}$ | KvP §2.1             |               |
| Tue 1 Sep  |                | Subgroups; the unit group $(\mathbb{Z}/n\mathbb{Z})^*$             | KvP §2.2             | Problem Set 1 |
| Thu 3 Sep  | <b>Q&amp;A</b> | Defense 1                                                          |                      |               |
| Tue 8 Sep  |                | Group homomorphisms; the order of an element                       | KvP §2.3             |               |
| Thu 10 Sep |                | Cyclic groups; primitive roots                                     | KvP §2.3             |               |
| Tue 15 Sep |                | Cosets and Lagrange's theorem                                      | KvP §2.4             | Problem Set 2 |
| Thu 17 Sep | <b>Q&amp;A</b> | Defense 2                                                          |                      |               |
| Tue 22 Sep |                | Normal subgroups and quotient groups                               | KvP §2.5             |               |
| Thu 24 Sep |                | The homomorphism theorem                                           | KvP §2.6             |               |
| Tue 29 Sep |                | Euler's theorem, fast modular exponentiation, and RSA end to end   | KvP App. B; KvP §2.4 |               |
| Thu 1 Oct  | <b>Exam</b>    | Midterm Exam 1 — Segment 1                                         |                      |               |
| Tue 6 Oct  |                | Rings and subrings                                                 | KvP §3.2             |               |
| Thu 8 Oct  |                | Ring homomorphisms                                                 | KvP §3.3             |               |
| Thu 15 Oct |                | Ideals                                                             | KvP §3.3             |               |
| Tue 20 Oct |                | Quotient rings                                                     | KvP §3.3             | Problem Set 3 |

| Date       |                | Topic                                                                                                                           | Reading                                                                                                                              | Due           |
|------------|----------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Thu 22 Oct | <b>Q&amp;A</b> | Defense 3                                                                                                                       |                                                                                                                                      |               |
| Tue 27 Oct |                | Fields and skew fields; $F_p$                                                                                                   | KvP §3.4                                                                                                                             |               |
| Thu 29 Oct |                | Integral domains and the field of fractions                                                                                     | KvP §3.5                                                                                                                             |               |
| Tue 3 Nov  |                | Unique factorization: UFDs                                                                                                      | KvP §3.7                                                                                                                             | Problem Set 4 |
| Thu 5 Nov  | <b>Q&amp;A</b> | Defense 4                                                                                                                       |                                                                                                                                      |               |
| Tue 10 Nov |                | PIDs and Euclidean domains                                                                                                      | KvP §3.7                                                                                                                             |               |
| Thu 12 Nov |                | Complex numbers: arithmetic, the plane, polar form, roots of unity                                                              | Beardon §3.1-3.2, §3.5; KvP §5.1-5.2                                                                                                 |               |
| Tue 17 Nov |                | Mobius transformations: the formula, composition as matrix multiplication, circles to circles, the Riemann sphere, fixed points | Beardon Ch. 13; Arnold–Rogness; Needham Ch. 3 §I-II, §IV-VI (handout; a copy is on reserve); Mobius Transformations Revealed (2 min) | Problem Set 5 |
| Thu 19 Nov | <b>Q&amp;A</b> | Defense 5                                                                                                                       |                                                                                                                                      |               |
| Tue 24 Nov |                | The Gaussian integers: norm, division with remainder, irreducibles, and sums of two squares — a survey                          | Conrad                                                                                                                               |               |
| Tue 1 Dec  | <i>Review</i>  | Review — the exam pool                                                                                                          |                                                                                                                                      |               |
| Thu 3 Dec  | <b>Exam</b>    | Midterm Exam 2 — Segment 2 weighted                                                                                             |                                                                                                                                      |               |